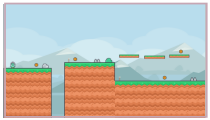
Player-1-Gym: Learning from Synthetic Experience

16 synthetic environments organized around shared control demands


2D shooting


2D platforming


2D circuit driving


3D aiming


3D road driving


3D chase

...

Environment randomization

Appearance
Color + lighting

Camera
View + projection

Scene
Layout + spawns

Control
Response gains

Controls

Screenshots

Verifiable rewards

Policy

Task instruction + current screenshot + interaction history

Rollout

Update

PPO on policy-generated experience

Screenshots + executed actions + verifiable rewards